

How We Study Multilingual Training in Theorem Proving

A simple overview of our pilot experiment

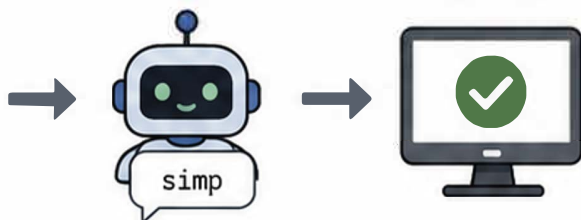
1 What is theorem proving?

1 Proof state

2 Model suggests a tactic

3 Proof assistant checks it

Goal:
 $\forall n : \mathbb{N}, n + 0 = n$



The model sees the current proof situation, suggests a next step, and the proof assistant checks whether it works.

2 Three training conditions

E1: Lean only



Monolingual
baseline

E3: Lean + Rocq



Real multilingual
training

E4: Lean + pseudo-Lean



Same Lean content, but
local names are renamed

```
h -> pseudo_1  
x -> pseudo_2
```

3 What do the comparisons mean?

E3 vs E1



Does adding a second
proof language help?

E3 vs E4



Is improvement caused by real
cross-language transfer, or just by
extra variation in the training data?

4 How we evaluate

A. Level 1: Next-step diagnostics



Exact match



Parseable output



Evaluation loss

B. Level 2: Proof-search evaluation



Pass@1



Pass@k



Compilable tactics



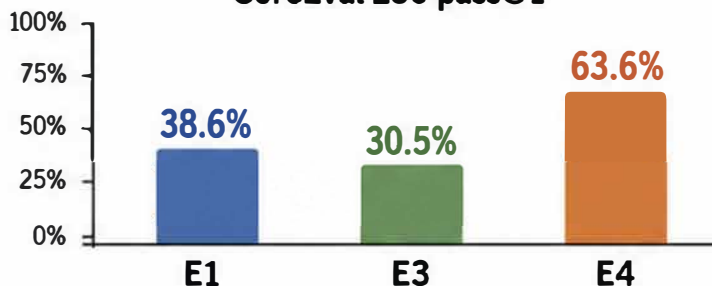
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Benchmark: CoreEval 250 and miniF2F



**Main
takeaway**

CoreEval 250 pass@1



In this pilot, **E4** performed best.

This suggests that extra training variation may help more than real cross-language transfer at this scale.



On a tiny miniF2F subset, **E3** showed a small advantage.